

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT – TEST CASE DESIGN ASSIGNMENT

Fraud Detection and Financial Transaction Monitoring System

Field	Details	Field	Details
Course Code	CSA10 / CSA1063	Course Title	Software Engineering
CO Assessed	CO4	Assessment	Test Case Design Assignment
Weightage	20%	Total Marks	25
CO4 Statement	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics	Duration	60 minutes
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Date	21/08/2026	Domain	Financial Technology / Fraud Monitoring

Assessment Approach and Model-Answer Method

This document follows the supplied model-answer method: identify functional and non-functional requirements, define test scenarios, design detailed test cases, apply test-design techniques, map requirements to tests, and summarize coverage. The original model uses the Digital Farmer Marketplace domain; here the same structure is adapted to the Fraud Detection and Financial Transaction Monitoring System.

Rubric Area	Excellent-level approach used
Requirement & Scenario Identification (15%)	Complete functional/non-functional requirements and comprehensive scenarios.
Test Case Design & Completeness (25%)	Normal, boundary, invalid and exceptional conditions.
Test Data & Expected Results (20%)	Specific data and precise expected outcomes.
Test Design Techniques (15%)	Equivalence Partitioning, Boundary Value Analysis, Decision Table and State Transition.
Traceability & Coverage (15%)	Requirement-to-test mapping and coverage summary.
Documentation & Presentation (10%)	Consistent IDs, headings, tables and professional format.

1. Functional and Non-Functional Requirements

1.1 Functional Requirements

ID	Requirement
FR01	Authorized users can log into the monitoring system.
FR02	Users can view financial transactions.
FR03	Transaction ID, amount, date/time, account/reference and status are displayed correctly.
FR04	Users can search/filter transactions.
FR05	Valid financial transactions can be submitted/recorded.
FR06	Transaction inputs are validated.
FR07	Transactions are analysed for suspicious behaviour.
FR08	A fraud risk score/risk level is calculated.
FR09	High-risk transactions generate alerts.
FR10	Authorized users can view/update alert status.
FR11	Frontend communicates correctly with backend APIs.
FR12	Backend returns valid transaction/fraud information.
FR13	Invalid/unavailable API requests are handled appropriately.
FR14	Transaction and alert history is maintained.
FR15	Dashboard presents monitoring/report information.

1.2 Non-Functional Requirements

ID	Requirement	Testing Focus
NFR01	Usability	Interface and messages should be easy to understand.
NFR02	Performance	Normal requests should respond within an acceptable time.
NFR03	Reliability	System should operate without unexpected crashes.
NFR04	Availability	Services should be accessible when running.
NFR05	Security	Unauthorized access and unsafe inputs should be handled safely.
NFR06	Maintainability	Modules should be understandable and testable.
NFR07	Scalability	System should support increasing transaction volume.

NFR08	Compatibility	Application should work in supported browsers.
NFR09	Portability	Containerized services should run consistently.
NFR10	Data Integrity	Transaction/alert records should not be corrupted or lost.

2. Test Scenarios

ID	Scenario	Objective
TS01	Application Access	Open monitoring application successfully.
TS02	Valid Login	Authenticate with valid credentials.
TS03	Invalid Login	Reject incorrect credentials safely.
TS04	Transaction Display	Display stored transaction details correctly.
TS05	Transaction Search/Filter	Find relevant transactions using search/filter.
TS06	Valid Transaction	Accept and store valid transaction data.
TS07	Invalid Transaction Input	Reject empty, malformed or invalid data.
TS08	Amount Boundary	Verify transaction amount boundary values.
TS09	Fraud Analysis	Analyse a transaction for suspicious behaviour.
TS10	Risk Scoring	Return risk score/risk level.
TS11	High-Risk Alert	Generate alert for high-risk transaction.
TS12	Alert Management	View/update alert status.
TS13	Frontend–Backend API	Verify correct API communication.
TS14	Invalid API/Error Handling	Handle invalid endpoint/service error safely.
TS15	Docker Deployment	Start services and verify communication.
TS16	Restart & Compatibility	Restart services and verify supported browser.

3. Test Design Techniques

3.1 Equivalence Partitioning

Input Class	Example	Expected
Valid positive amount	₹1,000	Accept
Zero amount	₹0	Reject

Negative amount	-₹500	Reject
Non-numeric amount	ABC	Reject
Empty amount	Blank	Reject

3.2 Boundary Value Analysis

Boundary	Example	Expected
Below minimum	-₹1	Reject
Minimum boundary	₹0	Reject if amount > 0
Just above minimum	₹1	Accept
Normal	₹1,000	Accept

3.3 Decision Table

Transaction	Risk	Authorized	Expected
Valid	Low/Medium	Yes	Accept and record
Valid	High	Yes	Record and generate alert
Invalid	Any	Yes	Reject
Valid	Any	No	Deny operation

3.4 State Transition Testing

Submitted → Validated → Analysed → Low/Medium Risk → Completed

Submitted → Validated → Analysed → High Risk → Alert Generated → Under Review → Resolved

Submitted → Validation Failed → Rejected

4. Detailed Test Cases

TC01 – Open Monitoring Application

Field	Details
Test Case ID	TC01
Test Scenario	Open Monitoring Application
Test Steps	Start services; open browser; enter URL.
Test Data	http://localhost:5173
Expected Result	Login/home page loads successfully.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC02 – Valid User Login

Field	Details
Test Case ID	TC02
Test Scenario	Valid User Login

Test Steps	Open login; enter valid credentials; click Login.
Test Data	admin / valid test password
Expected Result	User is authenticated and dashboard opens.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC03 – Invalid User Login

Field	Details
Test Case ID	TC03
Test Scenario	Invalid User Login
Test Steps	Enter incorrect credentials and submit.
Test Data	admin / wrong123
Expected Result	Login is rejected with safe error message.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC04 – Display Transaction

Field	Details
Test Case ID	TC04
Test Scenario	Display Transaction
Test Steps	Login; open Transactions; select record.
Test Data	TX1001
Expected Result	Transaction details are displayed correctly.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC05 – Search Transaction

Field	Details
Test Case ID	TC05
Test Scenario	Search Transaction
Test Steps	Enter transaction ID in search.
Test Data	TX1001
Expected Result	Matching transaction is displayed.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC06 – Valid Transaction

Field	Details
Test Case ID	TC06
Test Scenario	Valid Transaction
Test Steps	Enter valid transaction and submit.
Test Data	TX1002, ₹1,000, INR

Expected Result	Transaction is validated and stored.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC07 – Empty Mandatory Field

Field	Details
Test Case ID	TC07
Test Scenario	Empty Mandatory Field
Test Steps	Leave amount blank and submit.
Test Data	Blank amount
Expected Result	Validation message is displayed; record not created.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC08 – Zero Amount Boundary

Field	Details
Test Case ID	TC08
Test Scenario	Zero Amount Boundary
Test Steps	Enter zero amount and submit.
Test Data	₹0
Expected Result	Amount is rejected.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC09 – Negative Amount

Field	Details
Test Case ID	TC09
Test Scenario	Negative Amount
Test Steps	Enter negative amount and submit.
Test Data	-₹500
Expected Result	Negative amount is rejected.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC10 – Non-Numeric Amount

Field	Details
Test Case ID	TC10
Test Scenario	Non-Numeric Amount
Test Steps	Enter letters in amount and submit.
Test Data	ABC
Expected Result	Input is rejected; no record created.
Actual Result	To be verified during actual execution.

Pass/Fail Status	PENDING – update after execution.
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TC11 – Fraud Risk Analysis

Field	Details
Test Case ID	TC11
Test Scenario	Fraud Risk Analysis
Test Steps	Submit valid test transaction; run analysis.
Test Data	TX1003, ₹25,000, unusual test attributes
Expected Result	Fraud service returns risk score/level.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC12 – High-Risk Alert

Field	Details
Test Case ID	TC12
Test Scenario	High-Risk Alert
Test Steps	Submit high-risk test transaction; inspect alerts.
Test Data	High-risk test transaction
Expected Result	Alert is generated and linked to transaction.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC13 – Frontend–Backend API

Field	Details
Test Case ID	TC13
Test Scenario	Frontend–Backend API
Test Steps	Request transaction data from frontend.
Test Data	GET /api/transactions
Expected Result	Valid backend response is received and displayed.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC14 – Invalid API Request

Field	Details
Test Case ID	TC14
Test Scenario	Invalid API Request
Test Steps	Call an invalid endpoint.
Test Data	GET /api/invalid
Expected Result	Appropriate error response is returned; service does not crash.
Actual Result	To be verified during actual execution.

Pass/Fail Status	PENDING – update after execution.
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TC15 – Docker Startup

Field	Details
Test Case ID	TC15
Test Scenario	Docker Startup
Test Steps	Run Docker Compose; inspect services.
Test Data	docker compose up -d
Expected Result	Required services start and communicate.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC16 – Docker Restart

Field	Details
Test Case ID	TC16
Test Scenario	Docker Restart
Test Steps	Stop and restart services; reopen app.
Test Data	docker compose down / up -d
Expected Result	Services restart and application becomes available.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC17 – Browser Compatibility

Field	Details
Test Case ID	TC17
Test Scenario	Browser Compatibility
Test Steps	Open app in supported browser; use dashboard.
Test Data	Chrome / supported browser
Expected Result	Core pages load and operate correctly.
Actual Result	To be verified during actual execution.
Pass/Fail Status	PENDING – update after execution.

TC18 – Unauthorized API Access

Field	Details
Test Case ID	TC18
Test Scenario	Unauthorized API Access
Test Steps	Call protected endpoint without authentication.
Test Data	Protected endpoint without token
Expected Result	Access is denied safely.
Actual Result	To be verified during actual execution.

Pass/Fail Status	PENDING – update after execution.
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5. Requirement Traceability

Requirement	Covered Area	Test Cases
FR01	Login/access	TC01–TC03
FR02–FR03	Transaction display	TC04
FR04	Search/filter	TC05
FR05	Transaction recording	TC06
FR06	Validation	TC07–TC10
FR07–FR08	Fraud analysis/risk scoring	TC11
FR09	High-risk alert	TC12
FR10	Alert management	TC12; add status-update test
FR11–FR12	Frontend/API	TC13
FR13	Error handling	TC03, TC07–TC10, TC14
FR14	History	TC04, TC12; add persistence verification
FR15	Dashboard	TC04, TC13
NFR03–NFR04	Reliability/availability	TC15–TC16
NFR05	Security	TC03, TC18
NFR08	Compatibility	TC17
NFR09	Portability	TC15–TC16
NFR02	Performance	Dedicated response-time/load test required

6. Test Coverage Summary

Metric	Result / Plan
Total designed test cases	18
Normal/positive cases	TC01, TC02, TC04–TC06, TC11–TC13, TC15–TC17
Invalid/negative cases	TC03, TC07, TC09, TC10, TC14, TC18
Boundary tests	TC08–TC09
Equivalence Partitioning	TC07–TC10
Boundary Value Analysis	TC08–TC09
Decision Table	Transaction validity × risk × authorization
State Transition	Submitted → Validated → Analysed → Risk/Alert states
Functional coverage	Authentication, transactions, fraud analysis, alerts, API, deployment
Non-functional coverage	Security, reliability, availability, compatibility, portability; performance needs dedicated execution

7. Evidence / Screenshots to Attach

- Login page / monitoring dashboard
- Transaction list and transaction details
- Fraud risk score and high-risk alert
- Backend API response
- Docker Compose services running
- Test execution results

Use screenshots from the actual project execution. Do not claim an execution result unless it has been tested.

8. Conclusion

This test case design covers the major functional and non-functional requirements of the Fraud Detection and Financial Transaction Monitoring System. It includes normal, boundary, invalid and exceptional conditions and applies Equivalence Partitioning, Boundary Value Analysis, Decision Table and State Transition Testing.

The traceability matrix maps requirements to test cases and highlights remaining gaps. Before final submission, execute the application and replace the Actual Result and PENDING status fields with the observed results and attach the corresponding screenshots.